Lawrence Du, Ph.D.

Machine Learning Engineering • ML Platforms • MLOps
larrydu88@gmail.com | 626-808-7096 | Santa Clara, CA
github.com/LarsDu | linkedin.com/in/LarsDu | dublog.net

Skills

Languages: Python, SQL, C/C++, C#, Rust, Bash
ML & data: PyTorch, TensorFlow, Ray, MLflow, NumPy, pandas, scikit-learn
Infrastructure: AWS, Google Cloud, Kubernetes, Docker, Terraform, Pulumi, Metaflow
Methods: Foundation model fine-tuning, transformers, diffusion, single-cell analysis, RAG, MCP, graph community detection

Experience

GSK • Senior AI/ML Engineer, Sequence Learning and Phenomics
Oct 2024 – Present | South San Francisco, CA

- Built a sequence foundation model fine-tuning platform for RNA property and single-cell perturbation prediction, cutting iteration time **from weeks to less than a day**.
- Integrated Segment Anything Model (SAM) segmentation into production image-embedding pipelines for single-cell tracking and cell-state world modeling across **millions of images**.
- Designed embedding-based differentiation QC for **>3 million iPSC-to-microglia cells**, supporting Alzheimer's disease research.
- Piloted x-prediction / JiT diffusion models generating protein subcellular localization images from ESM protein embeddings.
- Built a retrieval-augmented LLM pipeline prioritizing **~1,000 candidate genes** for a lung-epithelium CRISPR knockout screen supporting a next-generation COPD drug program.
- Shipped MCP servers enabling company-wide LLM assistants to build and launch scientists' image-analysis pipelines, accelerating analysis turnaround **>10×**.

Freenome • Senior Machine Learning Research Engineer
Aug 2022 – Jun 2024 | South San Francisco, CA

- Led development of a distributed ML platform for cancer detection using PyTorch, Ray, and Kubernetes; accelerated training **>10×** with distributed data parallelism (DDP).
- Deployed organization-wide MLflow tracking on Google Cloud with Terraform and Pulumi; automated reproducible reporting and reduced researcher manual effort **≥5×**.
- Built multitask neural networks and elastic net models for colorectal cancer risk from cell-free DNA in a clinical trial cohort of **>27,000 individuals**.

23andMe • Machine Learning Engineer / Data Scientist
Nov 2018 – Aug 2022 | Sunnyvale, CA | Data Scientist prior to 2020

- Created and deployed Recent Ancestor Locations (RAL), a high-precision country-matching algorithm serving **>15 million customers**; advanced graph community detection for genetic subpopulations in Mexico and the UK.
- Built SNP feature engineering pipelines at **~10 million samples × ~1 million SNPs** using AWS Batch, Metaflow, Glue, and Athena for genome-wide association studies and polygenic risk scores (PRS).
- Improved type 2 diabetes and coronary artery disease PRS models using stacking ensembles; automated classifier reports with MLflow and Jupyter, reducing analysis time **from days to minutes**.

Scripps Research • Bioinformatician IV • May–Oct 2018
Developed an RNA-based transplant rejection classifier and Nanopore pipelines using Common Workflow Language.

Juno Diagnostics • Independent Consultant • Sep 2017 – Feb 2018
Developed a TensorFlow prenatal screening classifier and co-invented US20210020314A1: Deep learning-based prenatal testing.

Education

Ph.D., Biology • UC San Diego • 2010–2017 • Rifkin Lab; DeepNuc CNN sequence classification
B.A., Biological Sciences • Cornell University • 2006–2010 • Genetics and Development, *magna cum laude*

Selected Projects

Rogue Stargun — Unity/C# VR space simulator; AI flight planning, procedural missions, text-to-speech; 72 FPS on Quest 2, 90 FPS on Quest 3.
DiffuMon — Denoising diffusion probabilistic model (DDPM) image generator.